VINELAND

Move digital dollars *without exposing* who paid whom.

The post-quantum privacy layer for stablecoin payments on Solana, for businesses and AI agents.
The payment is public. The payer isn't.

ISLANDHACK V5 · FLORIANÓPOLIS · SOLANA DEVNET



Public rails publish the one thing a business cannot: who paid whom.

Stablecoins made dollars programmable, and Solana made them fast and cheap to move. Every transfer still prints the sender, the receiver and the pattern of a business on a ledger anyone can read, forever.

The fixes on offer hide the wrong thing. Confidential balances hide the amount, the one thing an auditor needs. Open privacy pools admit anyone, so a regulated desk cannot touch them. Plain transfers hide nothing.

A treasury paying suppliers in USDC

cannot let competitors rebuild its vendor book from public transfers.

A fund distributing to investors

cannot let the market read its investor list off the ledger.

A company running payroll on chain

cannot publish who it pays and how often.

Private who. Public how much.

Keep the amount, the time and the receipt public. Admit only approved wallets. Hide one link: which approved wallet a payout came from.

That is the split a regulated business can sign. The auditor keeps who came in and everything that moved. The market learns nothing about who paid whom.

Public	Private
Which approved wallets deposited	Which deposit a payout came from
Every amount, one fixed note size	The depositor’s secret
Every payout, its recipient, its time	Whether two payouts share a payer
The membership root, spent nullifiers	The link from commitment to nullifier

One privacy layer. Approved wallets in, clean wallets out.

01 · APPROVE

The operator approves a wallet, a company's or an AI agent's, after KYC. The approval is an account on chain: checkable by anyone, required to deposit.

02 · DEPOSIT

The approved wallet moves one fixed-size note of a USDC-style token into a vault the program controls and publishes a commitment, a hash of a secret it keeps.

03 · WITHDRAW

The depositor proves membership on its own device. Solana verifies the proof on chain. A relayer submits the payout to a brand-new wallet; the depositor's key never signs it.

A public way out: revealing the note's secret returns the full note to the depositor, with no signature from anyone, and burns the same nullifier a withdrawal would, so a note leaves once, privately or publicly, never both.

The payment is on the ledger. The payer is not.

THE PRIVATE WITHDRAWAL `Sg7NfZ2z…J59t9Cw on devnet`

Field	In the run
Signer	the relayer, alone
Clean wallet	980,300 of a 1,000,000 note
Treasury	19,700, the 1.97% fee
Depositor’s key	absent
Authorized by	a STARK the chain verified, no committee

REFUSED ON CHAIN

Attempt	Code
Deposit without approval	0xbc4
Withdraw before verified	0x1771
Relayer swaps the recipient	0x1772
Replay a withdrawal	0x0, spent
Withdraw, then exit too	0x0, spent
Exit with another’s secret	0x1778

Solana devnet, no committee: pool 3W1FD5Q1…5VXi, verifier 8vPHUTH2…kcYo. 239 transactions, 213 of them on-chain verification; every signature in docs/DEVNET_RUN_COMMITTEE_FREE.md. Devnet only.

A ledger never forgets. So the privacy cannot expire.

HASH-ONLY RECORD

Nothing for Shor

Every on-chain value that relates a payout to a deposit is a hash: commitment, nullifier, root. Harvest the ledger today, run a quantum computer later, and there is no curve to break.

NO TRUSTED SETUP

No ceremony

riverrun proofs are Circle STARKs over Mersenne-31 with Poseidon2. There is no secret parameter whose leak would forge proofs or unmask users.

MEASURED, NOT ADVERTISED

30 vs 1.0 to 6.5

On a live Solana mainnet pool advertising 30 members, riverrun's ruler measured an effective set between 1.0 and 6.5. We measure ours the same way.

Precisely: everything that authorizes a payout is a hash, the commitment, the nullifier, the on-chain tree, and the STARK Solana verifies. What stays classical is Solana's own transaction signatures, which cannot redirect or forge a payout.

No signer can approve a payout. Here is exactly what is still trusted.

What it trusts

The two programs, the pool and riverrun's verifier, and their upgrade authorities (devnet: the author's key). **The operator, for the allowlist:** it decides who may deposit. It cannot publish a root, approve a payout or block an exit.

What it does not trust

Soundness: the proofs are verified on chain and the root is computed on chain. **Privacy:** the proof hides which deposit it was. **Custody:** the vault belongs to the program. **The relayer:** the recipient is bound into the proofs.

On devnet, the depositor proved in 5.0 s on a laptop; riverrun's verifier program checked both proofs in 213 transactions, the heaviest at 1.21M of the 1.4M compute cap; the pool paid from the verified buffer in 25,836 CU. Solana's 4 KB transactions would cut the upload about fourfold.

1.97%, once, on the private payout.

PROTOCOL FEE

1.97%

of each private payout, paid from the note to the treasury in the same transaction. One rate whether the note is small or large.

EXIT

free

Ragequit returns the full note to the depositor. Deposits carry no protocol fee. The fee is for the private payout, not for holding money.

CUSTODY

none

The vault is controlled by the program. The fee is computed on chain, rounded down, and capped in code.

In the devnet run: a 1,000,000-unit note paid 980,300 to the clean wallet and 19,700 to the treasury. On a 100 USDC note that is 98.03 USDC and a 1.97 USDC fee.

Don’t take our word for it.

STELLAR MAINNET

true

a post-quantum membership proof verified on chain, 11 Aug 2026. Contract CB32KP47…DXW3L6, tx 9f776be5…5084.

SOLANA DEVNET

18

riverrun actions settled in one transaction, after a full 8-member round with the floor and anti-replay enforced on chain.

SOLANA DEVNET

213 tx

both proofs verified on chain by riverrun’s verifier program, then the payout, with no committee key anywhere in the run.

MEASURED ANONYMITY

1.0 to 6.5

effective set the ruler found behind an advertised 30 on a live Solana mainnet pool.

Built in the window: the Solana program and client, 18 program tests and 2 end-to-end tests with real proofs passing, the committee-free flow on Solana devnet with the proofs verified on chain, a relayer service and a CLI. Also: Stellar Ambassador, Brazil chapter. Two merged pull requests to Nethermind’s Stellar Private Payments, one a soundness fix. Built without external capital. Receipts at **vineland.app/proofs**.

Built in Santa Catarina.
Ready to be broken.

Manuel Guilherme, founder and engineer. Security across web2 and web3, builder of riverrun, Stellar Ambassador.

The ask. One pilot partner with a real payout flow to run on devnet. Reviewers who will attack the invariants in our results dossier. Introductions toward an external audit, the gate before any mainnet.

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1.97% ON THE PRIVATE PAYOUT · NON-CUSTODIAL · POST-QUANTUM PRIVACY RECORD

